

# p-Block Elements

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

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## 1. Notes

- The p-block consists of groups 13-18, where the last electron enters an np orbital; includes metals, metalloids, and non-metals.
- General trend: metallic character decreases and non-metallic character increases across a period (left to right); metallic character increases down a group.
- Group 13 (Boron family): boron shows anomalous behavior (small size, high ionization energy) compared to other members; aluminum is amphoteric.
- Group 14 (Carbon family): carbon uniquely forms strong pi bonds and long chains (catenation); shows allotropes like diamond and graphite.
- Group 15 (Nitrogen family): nitrogen exists as a stable diatomic gas (N triple-bond N); phosphorus shows allotropy (white, red phosphorus).
- Group 16 (Oxygen family): oxygen is highly electronegative and forms strong hydrogen bonds; sulfur shows extensive allotropy and catenation.
- Group 17 (Halogens): highly reactive non-metals, strong oxidizing agents; oxidizing power decreases down the group ( $F_2 > Cl_2 > Br_2 > I_2$ ).
- Group 18 (Noble gases): have complete octets (except He, which has 2 electrons), making them largely chemically inert; only heavier noble gases (Xe, Kr) form a limited number of compounds.

|          |          |          |          |          |           |
|----------|----------|----------|----------|----------|-----------|
| <b>B</b> | <b>C</b> | <b>N</b> | <b>O</b> | <b>F</b> | <b>Ne</b> |
| 13       | 14       | 15       | 16       | 17       | 18        |

p-block: groups 13-18, filling np orbitals

*The p-block of the periodic table: groups 13 to 18, where the np orbitals are being filled.*

## 2. Formula Sheet

|                                              |                                                                         |
|----------------------------------------------|-------------------------------------------------------------------------|
| General valence configuration (p-block)      | $ns^2 np^{(1 \text{ to } 6)}$                                           |
| Group 13 general oxidation state             | +3 (also +1 for heavier elements, inert pair effect)                    |
| Group 15 general oxidation states            | -3, +3, +5                                                              |
| Group 16 general oxidation states            | -2, +2, +4, +6                                                          |
| Group 17 general oxidation states            | -1 (also +1, +3, +5, +7 for Cl, Br, I with O/F)                         |
| Inert pair effect (qualitative)              | Heavier p-block elements favor oxidation state 2 less than group number |
| Oxidizing power of halogens                  | $F_2 > Cl_2 > Br_2 > I_2$                                               |
| Acidic strength of hydrides (group 15/16/17) | Increases down the group                                                |

## 3. Practice Problems (22 questions)

1. State the general electronic configuration of p-block elements.
2. Explain why boron shows anomalous properties compared to other group 13 elements.
3. State why aluminum is called an amphoteric element, with one example reaction each with an acid and a base.
4. Explain the term 'catenation' and state which p-block element shows it most extensively.
5. State the two most important allotropes of carbon and one key structural difference between them.
6. Explain why nitrogen ( $N_2$ ) is chemically much less reactive than phosphorus, despite being in the same group.
7. State the trend in oxidizing power of the halogens from  $F_2$  to  $I_2$ , and explain the reason for this trend.
8. Explain why noble gases are generally unreactive, and name one noble gas that forms compounds.
9. State the inert pair effect and explain how it affects the stability of lower oxidation states in heavier p-block elements (e.g., Tl or Pb).
10. Explain why HF has a much higher boiling point than HCl, despite fluorine being lighter than chlorine.
11. State the general trend of acidic strength of the hydrogen halides (HF to HI) down group 17.
12. Explain why oxygen commonly shows a -2 oxidation state but rarely +2, in terms of its high electronegativity.
13. State one major industrial use of sulfuric acid ( $H_2SO_4$ ) or a p-block element/compound of your choice.
14. Explain why silicon, unlike carbon, does not readily form long chains (catenation) to the same extent.
15. State the trend in atomic size across the p-block from group 13 to group 18 within the same period.

16. Explain why chlorine is a stronger oxidizing agent than iodine, in terms of electron affinity and bond dissociation energy trends.
  17. State two uses of noble gases (e.g., helium, neon, argon) in daily life or industry.
  18. Explain why the +5 oxidation state becomes less stable while +3 becomes more stable going down group 15 (inert pair effect).
  19. State the general formula and bonding nature of oxides formed by p-block elements across a period (basic to acidic trend).
  20. Explain why phosphorus shows allotropy (white and red phosphorus) with very different reactivity.
  21. State why fluorine does not show a positive oxidation state, unlike other halogens.
  22. Explain the term 'metalloid' and give two examples of elements from the p-block classified as metalloids.
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## Answer Key

1.  $ns^2 np^{(1 \text{ to } 6)}$
2. Conceptual - boron's small size and high ionization energy give it more covalent character and different chemistry compared to Al, Ga, In, Tl
3. Conceptual - amphoteric means it reacts with both acids and bases; e.g.,  $Al + HCl \rightarrow AlCl_3 + H_2$  (with acid),  $Al + NaOH \rightarrow NaAlO_2 + H_2$  (with base)
4. Conceptual - catenation is the ability of atoms of the same element to form chains/rings via self-linking bonds; carbon shows this most extensively
5. Conceptual - diamond (each C bonded to 4 others in a tetrahedral 3D network, very hard) vs graphite (layered hexagonal sheets with delocalized electrons, soft and conducting)
6. Conceptual -  $N_2$  has a very strong triple bond (N triple-bond N) with high bond dissociation energy, making it kinetically very stable/unreactive at room temperature
7. Conceptual - oxidizing power decreases  $F_2 > Cl_2 > Br_2 > I_2$ , due to decreasing electronegativity and weaker attraction for electrons as atomic size increases down the group
8. Conceptual - noble gases have a complete stable octet (or duplet for He), giving little tendency to gain, lose, or share electrons; Xe forms compounds like  $XeF_2$ ,  $XeF_4$
9. Conceptual - inert pair effect: the  $ns^2$  electron pair becomes increasingly reluctant to participate in bonding for heavier elements, favoring an oxidation state 2 less than the group's maximum (e.g.,  $Tl^+$  more stable than  $Tl^{3+}$ )
10. Conceptual - HF molecules form strong hydrogen bonds (F is highly electronegative and small), requiring more energy to separate, raising its boiling point unusually high
11. Acidic strength increases:  $HF < HCl < HBr < HI$  (down the group, due to decreasing bond strength/increasing bond length)
12. Conceptual - oxygen's very high electronegativity strongly favors gaining electrons (-2 state); achieving a +2 state would require it to lose electrons, which is energetically unfavorable
13. Conceptual - e.g.,  $H_2SO_4$  widely used in fertilizer production, and various other industrial processes
14. Conceptual - silicon's larger atomic size gives weaker, longer Si-Si bonds compared to C-C bonds, making catenation less favorable energetically
15. Conceptual - atomic size generally decreases from group 13 to 18 across the same period, due to increasing nuclear charge
16. Conceptual - chlorine has a higher electron affinity and forms a stronger, shorter bond compared to iodine, making it a more effective oxidizing agent

**17.** Conceptual - e.g., helium in balloons/airships (low density, inert, safe); neon in signage/lighting; argon as an inert atmosphere for welding

**19.** Conceptual - oxides trend from basic (left, metals) to amphoteric (middle) to acidic (right, non-metals) across a period

**21.** Conceptual - fluorine is the most electronegative element, so it always gains electrons (only -1 state), never loses them to show a positive oxidation state

**18.** Conceptual - inert pair effect makes the  $ns^2$  electrons less available for bonding down the group, so +3 (using only np electrons) becomes relatively more stable than +5 (using ns and np electrons) for heavier group 15 elements

**20.** Conceptual - white phosphorus ( $P_4$  tetrahedral molecules, reactive, toxic) vs red phosphorus (polymeric structure, more stable, less reactive) - different structures give very different properties

**22.** Conceptual - metalloid has properties intermediate between metals and non-metals; e.g., silicon and germanium (or boron, arsenic, antimony)

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.